Admissibility Defines Structural Limits on Optimisation in Artificial Intelligence

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Abstract

Many persistent failures of contemporary artificial intelligence are treated as failures of optimisation, data coverage, or statistical robustness. This paper argues that a recurring class of these failures arises earlier: the observation regime under which a system acts may fail to preserve the distinctions required by the predicate it is asked to decide. We make this condition precise. A predicate $\Phi : X \to \{0, 1\}$ is decidable from observations $M : X \to O$ only when Φ factors through the quotient that M induces on X. This is an elementary factorisation condition. Its role here is not to supply a new theorem of set theory, but to state a prior proof obligation for consequential artificial intelligence systems.

The paper develops that obligation in four steps. First, it gives a finite worked example, with explicit matrices and an explicit residue, in which local coherence does not decide global assembly. The residual obstruction is called warrant debt. Second, it gives an executable finite certificate prototype in which a cyclic diagnostic system receives one of three accountable verdicts: coherence failure, global admissibility, or warrant debt. Third, it gives two artificial intelligence schemata, reward hacking and benchmark or proxy failure, each as an X, M, Φ triple. Fourth, it states an architecture in which productive search is separated from closure authority. The productive layer proposes, predicts, or optimises; the closure layer checks admissibility, reports inadequacy, and, where the regime fails, forces rejection or escalation. The paper also records the limits of the proposal, especially the unresolved problem of detecting inadmissibility when the effective observation map is implicit in learned weights rather than declared in the architecture. A final, speculative section treats apparent human randomness as possible evidence of regime change, but only as philosophical motivation for the architectural distinction.

Keywords: admissibility, warrant debt, observation partitions, factorisation, closure, descent, Goodhart's law, reward hacking, benchmark failure, formal verification, AI safety, boundary revision.

1 Introduction: From Optimisation Failure to Admissibility Failure

A neural classifier achieves state of the art accuracy on its benchmark and then misclassifies an image of a panda as a gibbon when a barely perceptible perturbation is added to the input (Goodfellow et al., 2015). A large language model performs better as it is scaled on most tasks but worse on a small family of tasks where success requires the model to override a surface pattern in its training distribution (McKenzie et al., 2023). A reinforcement learning agent achieves a perfect reward score by exploiting an unintended regularity of its environment that the designer did not anticipate (Amodei et al., 2016; Manheim and Garrabrant, 2018; Skalse et al., 2022). A deployed model trained on one population degrades sharply when applied to another that is, by every observable measure available to the engineers, similar (Hendrycks and Gimpel, 2017). None of these failures is unfamiliar, and none is repaired by additional training of the same kind that produced the system in the first place.

These examples are not claimed to share one empirical cause. Adversarial sensitivity, inverse scaling, reward hacking, and deployment shift arise from different mechanisms and have been studied by separate technical literatures. The claim here is narrower: these failures share a structural diagnostic form whenever the target distinction is not preserved by the observational regime through which the system acts. The point is not to flatten the technical differences among the cases, but to identify a condition that is logically prior to optimisation in all of them.

Before asking whether a system can optimise a target, we must ask whether the target is decidable under the system's observational regime. This is the paper's central question. Common remedies for artificial intelligence failure are optimisation remedies: more data, broader coverage, finer regularisation, a better-shaped reward signal, or a larger parameter count. Each has its place. None addresses the failure considered here. If the system is asked to make a distinction that its observational structure has already collapsed, optimisation is not the missing step. The question has been made ill-posed inside the regime.

The formal result used in this paper is elementary. It is the familiar descent condition for a predicate along a quotient induced by an observation map, and readers trained in algebra, logic, or theoretical computer science will recognise it as the universal property of a quotient set. The contribution is not a new theorem of set theory. The contribution is to place this condition where Dijkstra would have placed a proof obligation: before the computation whose result is to be trusted. The worked examples show what follows when that obligation is not discharged.

The paper makes five contributions. First, a formal admissibility condition for judgement under observation, together with the elementary proposition that inadmissibility cannot be optimised away. Second, a finite worked obstruction, built from a regional structure of local checks and global assembly, that shows concretely how local adequacy can fail to lift to global warrant, and that introduces the residual quantity we call warrant debt. Third, an executable finite certificate prototype, written for a cyclic diagnostic setting, that computes the obstruction, emits a certificate, and distinguishes three cases: local incoherence, global admissibility, and coherent local data with non-zero warrant debt. Fourth, two explicit artificial intelligence schemata, reward hacking and benchmark or proxy failure, each written out as an X, M, Φ triple and shown to instantiate the same obstruction structure as the finite case. Fifth, an architectural proposal that separates productive search from closure authority, with closure authority responsible for admissibility certification, inadequacy reporting, and, where warranted, controlled revision of the operative regime.

The remainder of the paper proceeds as follows. Section 2 introduces admissibility conceptually, without symbols. Section 3 distinguishes productive systems from closure-capable systems. Section 4 states the formal admissibility condition and proves the fixed-regime limit on which the rest of the argument depends. Section 5 gives the finite worked obstruction that supplies the missing concrete bridge between the elementary theorem and the artificial intelligence claims that follow. Sections 6 and 7 apply the framework to reward hacking and to benchmark or proxy failure, each as an explicit triple. Section 8 reinterprets Goodhart's law as the dynamic expression of the same structural failure. Section 9 treats boundary revision and apparent human randomness as philosophical motivation rather than as an established empirical claim. Section 10 draws the structural contrast between artificial and natural intelligence. Section 11 develops closure authority, warrant debt, certificate architecture, and the executable cyclic diagnostic prototype. Section 12 situates the paper among adjacent literatures. Section 13 sketches the generalisation of the deterministic theory to stochastic observation. Section 14 states the limitations and open problems of the proposal explicitly. Section 15 concludes. Appendix A proves the factorisation theorem in full and records its correspondence with the weakest-precondition calculus for non-deterministic guarded conditionals. Appendix B gives the technical detail of the finite regional obstruction. Appendix C describes the certificate schema underlying the architectural proposal. Appendix D sketches the stochastic generalisation.

2 What Judgement Requires

A system cannot distinguish what its observations collapse. If two situations look the same to the system, no downstream reasoning can separate them, regardless of how sophisticated the reasoning is. This is not a statement about computational power. It is a statement about the precondition for a question to be well-posed under a given observation.

A representational regime, however constructed, partitions the situations the system might encounter into groups that the regime treats as equivalent. Two situations fall into the same group when the regime cannot tell them apart. A judgement that the system might be asked to make, cast as a yes or no question about the situation, can be settled by the system only when the question respects this partition, that is, only when the answer is the same for any two situations the regime cannot distinguish. Where this fails, the question is well-posed in the world but malformed inside the system. We call this condition admissibility, and we say that the question is admissible with respect to the regime when the condition is met.

The point is easier to see in everyday terms before it is made formal, and the two examples used throughout this paper are chosen because they already anticipate the structure that will later be written symbolically. Consider a doctor equipped only with a thermometer. The state space X is the set of bodily states a patient might be in, the observation map M is the temperature reading the thermometer returns, and the predicate of interest Φ is a question such as whether the patient has an infection of a particular type. The thermometer can settle the question of fever. It cannot settle the question of which infection caused it, because two patients with different infections may share the same temperature reading, and no degree of additional skill in reading the thermometer recovers a distinction the instrument never preserved. Consider next a searcher equipped only with a metal detector. Here X is the set of buried-object states, M is the metallic or non-metallic signal the detector returns, and Φ is the predicate of interest, for instance whether the buried object is gold. The detector can settle whether something metallic is present. It cannot settle whether the metal is gold, because objects of differing value can share an identical metallic signature. In both cases the failure is not a failure of intelligence on the part of the doctor or the searcher. It is a limit on what the available observational instrument is in a position to be asked, and a more sophisticated decision procedure operating on the same instrument does not extend the range of questions the instrument can settle.

The artificial intelligence cases that motivate this paper have the same structure, but they are harder to diagnose because the observation maps involved are often learned, implicit, and high-dimensional, rather than declared in advance as a thermometer or a detector declares its own scope. A classifier's input channel may or may not preserve the distinction between an unperturbed image and an adversarially perturbed one whose underlying object differs in the relevant sense. A reward signal may or may not preserve the distinction between the configurations a designer intended to reward and configurations that exhibit the same scalar reward without the underlying property. A benchmark may or may not preserve the distinction between systems that are only benchmark-adequate and systems that are deployment-adequate. The structural question is the same across every case: does the representation preserve the distinctions required by the predicate being optimised? Sections 6 and 7 answer this question concretely for two of these cases. The present section has only set out what the question means.

A system that operates only by optimisation within a fixed regime has, by the construction of its own operations, no means of detecting that admissibility has failed. The detection of admissibility failure is a different operation from the optimisation of judgements within an admissible regime. We call this second operation closure, and we develop the concept in the next section.

3 Productive Systems and Closure Authority

Call a system productive when its repertoire of operations consists of search, ranking, prediction, generation, reward maximisation, or optimisation, conducted within a given observational regime. Classifiers, generative models, reinforcement learning agents, benchmark optimisers, search procedures, and automated tactic generators in proof assistants are all productive systems in this sense. Productive systems can be exceedingly capable, and nothing in this paper questions that capability. The claim is narrower: productive systems cannot, by those operations alone, certify the adequacy of the regime under which they operate.

Closure is the authority to determine whether the regime under which judgement is being made preserves the distinctions required for that judgement. We decompose closure into three components, each of a different character. The first is admissibility checking, the determination of whether a candidate predicate Φ descends along the canonical projection induced by the observation map M in force. The second is inadequacy detection, the reporting of signals that the current regime collapses distinctions relevant to the intended judgement, even before a clean formal proof of inadmissibility is available. The third is regime revision authority, the capacity, located either in the system itself or in a surrounding authority, to revise M when admissibility has failed.

Only the first of these three components is fully formalised in this paper, by the factorisation theorem of Section 4. The second and third are identified here as architectural obligations, and Section 5 supplies a finite prototype in which all three are visible in a tractable setting. Their general solution for learned neural systems, where the operative observation map is implicit in distributed weights rather than declared, remains open, and we state this in Section 14 rather than implying completeness.

The proof-assistant analogy is useful only if it is kept narrow. Proof assistants separate proposal from certification. Tactics propose derivations; a small kernel checks them, and no proof is admitted unless the kernel accepts the resulting term. Most kernel rejections concern ordinary proof failure: an ill-typed term, a malformed inductive definition, or a step that does not follow from the rules. These are not failures of admissibility in the broad sense used here. They are failures inside a declared calculus.

The analogy supports one claim: productive search and closure authority must be separated. It does not support the stronger claim that proof assistants reconstruct their own foundations. They do not. When a foundation is judged inadequate, revision is performed externally by the mathematical and engineering communities that maintain the formal system. Section 11 uses the analogy in this restricted sense: a productive component may propose; a closure component must check; reconstruction, when needed, belongs to an authority outside the productive search itself.

4 The Formal Core: Admissibility and Fixed-Regime Limits

We now make the foregoing precise. Let X be a state space and O a space of observations available to a system. Let

$$M : X \to O$$

be the observation map characterising the regime under which the system represents its environment. The map M induces an equivalence relation on X, namely

$$x \sim_M y \iff M(x) = M(y).$$

Two states are observationally indistinguishable when they are mapped to the same observation. The partition $X/\sim_M$ is the representational partition induced by the regime.

Definition 1 (Admissibility)**.** *A predicate $\Phi : X \to \{0,1\}$ is admissible with respect to the observation map M if and only if*

$$M(x) = M(y) \implies \Phi(x) = \Phi(y)$$

for all $x, y \in X$. Equivalently, Φ factors through M, so that there exists a predicate $\bar{\Phi} : O \to \{0,1\}$ with $\Phi = \bar{\Phi} \circ M$.

Proposition 1 (Fixed-Regime Limit)**.** *Let $M : X \to O$ be an observation map and $\Phi : X \to \{0,1\}$ a predicate that is not admissible with respect to M. Then no decision rule of the form $\delta : O \to \{0,1\}$ agrees with Φ on X. In particular, for any optimisation criterion defined on functions of O alone, the predicate Φ cannot be recovered.*

Proof. By hypothesis there exist $x, y \in X$ with $M(x) = M(y)$ but $\Phi(x) \neq \Phi(y)$. For any $\delta : O \to \{0,1\}$, the values $\delta(M(x))$ and $\delta(M(y))$ are equal. Hence $\delta \circ M$ cannot agree with Φ on both x and y. Since this holds independently of any criterion used to select δ, no optimisation over the family $\{\delta : O \to \{0,1\}\}$ produces agreement on X. □

The proposition is not technically deep. Its importance lies in what it forbids. No optimiser operating only on O can recover distinctions erased by M, however the optimiser is trained, scaled, or regularised. The issue is not insufficient search. The issue is malformed judgement under the standing regime, a defect that lies upstream of inference and that no operation downstream of M can repair.

Proposition 2 (Non-recoverability under post-processing)**.** *Let $M : X \to O$ be an observation map and let $\Phi : X \to \{0,1\}$ be a predicate not admissible with respect to M. Let $F : O \to B$ be any post-processing map. Then there is no decision rule $G : B \to \{0,1\}$ such that*

$$\Phi = G \circ F \circ M.$$

Proof. If such a G existed, then Φ would factor through M via $G \circ F : O \to \{0,1\}$, contradicting the failure of admissibility. □

Remark. Proposition 2 is the formal version of the claim that downstream processing cannot restore a distinction erased upstream. Calibration, reranking, decoding, benchmark aggregation, reward shaping, or any other operation performed only on the observed representation may improve performance when the required distinction is still present in the observation. It cannot recover a predicate whose relevant distinction has already been identified by the observation map. Repair of that failure requires either a refined observation regime or a certificate-bearing closure regime of the kind developed in Sections 5 and 11.

Written in Dijkstra's idiom, admissibility is the weakest precondition for asking a postcondition through an observation. The command "observe by M" identifies all states in the same fibre of M. A postcondition Φ may be checked after that command only when it is invariant under the identification the command performs. If it is not invariant, no later command over the observation can make it invariant. This is the calculational reason for separating construction from checking throughout the rest of the paper.

The content of Definition 1 is the recognition that not every predicate is sayable in a given representational regime. A predicate that distinguishes states which the regime cannot distinguish is, from the standpoint of that regime, malformed. No amount of subsequent optimisation, training, or fine-tuning within the same observational partition repairs the situation, because the problem is not that the system is ignorant of Φ but that, given M, the question Φ poses cannot be addressed at all. Whether a system is then permitted to recognise this failure is a matter of architecture, and it is to that question, by way of a worked example, that we now turn.

The next section gives a finite obstruction example in which the failure is witnessed by a computable residue.

5 A Finite Obstruction Example: Local Adequacy and Global Warrant Debt

Because the factorisation condition of Section 4 is completely general, its diagnostic value depends on showing concretely how admissibility failure appears in a structured system in which local observations appear adequate while a global predicate remains undecided. This section gives such an example. The example is finite and modest by design. It is not offered as a model of any specific deployed artificial intelligence system. It is a toy model that can be checked by hand. Its purpose is to show, rather than assert, that local success under a declared regime can coexist with global inadmissibility.

5.1 A finite regional system

Consider a finite collection of regions $U_1, \ldots, U_n$ covering a domain of interest, together with their pairwise overlaps $U_i \cap U_j$. Suppose that on each region U_i a local assignment a_i has been made, for instance a locally proposed value, label, or partial solution, and that on each overlap $U_i \cap U_j$ a comparison is recorded between the restrictions of a_i and a_j to the shared region. This comparison need not report agreement: it may instead record a discrepancy, a number measuring how far a_i and a_j diverge on the overlap. We reserve the term *local agreement* for the special case in which every recorded discrepancy is zero, and we use the weaker term *local profile* for the comparison data in general, whatever its values. A natural global question then arises: do the local assignments a_i assemble into a single object defined consistently across the whole domain? We call an assignment of this kind *globally coherent* when such an assembly exists. We say a local profile is *locally coherent* when it contains no contradiction across any triple overlap where three regions meet, a condition made precise below as $\delta_1 r = 0$. Local coherence is strictly weaker than local agreement, since a profile can be locally coherent while every overlap reports a non-zero discrepancy, provided those discrepancies are mutually consistent around each triple overlap where they meet. The point of the example is that local coherence and global coherence are not the same, and that the gap between them, a profile that is locally coherent yet not globally assemblable, is the kind of structural failure this paper is about.

5.2 The state space, the observation maps, and the predicate

Let X be the set of finite regional configurations, where a configuration $x \in X$ consists of a choice of regions, local assignments a_i on each region, and the overlap data comparing assignments on shared boundaries. Two observation maps are relevant to the example, and keeping them apart is essential to what the example shows.

The first is the local observation map

$$M_{\mathrm{loc}}(x) = \text{the local coherence report of } x,$$

recording what is visible to an observer restricted to local inspection: which overlap comparisons are present, and whether the resulting profile passes the local coherence check $\delta_1 r_x = 0$ at every triple overlap. This is a productive regime in the sense of Section 3: it records local data and local checks, but it does not itself perform, or report the result of, the exactness calculation that would decide whether the local profile is removable by correction.

The second is the certificate observation map

$$M_{\mathrm{cert}}(x) = (\delta_0, r_x),$$

where δ_0 is the correction operator determined by the regional structure and r_x is the residue associated with the configuration. This richer regime contains the data a closure layer needs to decide whether the residue is exact, and hence whether x is globally coherent. Constructing

M_{cert} requires nothing beyond what is already implicit in the regional structure; the point of distinguishing it from M_{loc} is that a system restricted to local inspection does not automatically have access to it.

Let the predicate of interest be

$$\Phi(x) = 1 \iff \text{the local data of } x \text{ admit a globally coherent assembly.}$$

The substance of the example, developed over the remainder of this section, is that Φ need not descend along M_{loc}, even though it does descend along M_{cert} once the certificate data are supplied and checked. The passage from M_{loc} to M_{cert} is the passage from productive local inspection to closure authority discussed in Section 3, made concrete enough here to verify by hand.

5.3 The residue and the obstruction

The technical content of the example, given in full in Appendix B, is that the gap between local coherence and global coherence can be captured by a residue r_x associated with each configuration, together with two operators familiar from the Čech-theoretic treatment of gluing problems: a local correction operator δ_0, which records the effect of adjusting local assignments, and a coherence operator δ_1, which records whether a given residue is itself internally consistent across triple overlaps. The configuration is globally coherent when

$$\Phi(x) = 1 \iff r_x \in \operatorname{im} \delta_0,$$

that is, when the residue can be cancelled by some local correction. The obstruction case, exhibited explicitly below and in full detail in Appendix B, is a configuration for which

$$\delta_1 r_x = 0 \quad \text{but} \quad r_x \notin \operatorname{im} \delta_0.$$

The first condition says that the residue is itself coherent, in the sense that it does not contradict itself across the triple overlaps where three regions meet, and is the condition recorded by M_{loc}. The second says that, despite this internal coherence, no choice of local correction removes it, a fact that M_{loc} does not record and that only M_{cert} makes decidable. Such a configuration presents a locally coherent profile to M_{loc}, with no contradiction visible to any local coherence check, and yet $\Phi(x) = 0$: no globally coherent assembly exists. The defect is not that a correction has not yet been found by further search. The defect is that no correction exists, a fact established once and for all by the same finite calculation that exhibits the residue, and that calculation is what M_{loc} withholds and M_{cert} supplies.

5.4 A worked numerical instance

The schema above is given concrete content by the following minimal example, with the full derivation in Appendix B. Take four regions U_1, U_2, U_3, U_4 arranged cyclically around a domain shaped like a ring, so that each region overlaps only its two neighbours: $U_1 \cap U_2$, $U_2 \cap U_3$, $U_3 \cap U_4$, and $U_4 \cap U_1$ are non-empty, while $U_1 \cap U_3$ and $U_2 \cap U_4$ are empty, and no three regions share a common point. A local correction assigns a number $c_i \in \mathbb{Q}$ to each region U_i, so $C^0 \cong \mathbb{Q}^4$. An overlap discrepancy assigns a number to each of the four overlaps, so $C^1 \cong \mathbb{Q}^4$, with coordinates ordered $(a_{12}, a_{23}, a_{34}, a_{41})$. Because no three regions overlap, there is no triple-overlap data to record, and $C^2 = 0$.

The local correction operator is

$$\delta_0 \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{pmatrix} = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \\ 1 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{pmatrix} = \begin{pmatrix} c_2 - c_1 \\ c_3 - c_2 \\ c_4 - c_3 \\ c_1 - c_4 \end{pmatrix},$$

recording, on each overlap, the discrepancy a proposed correction would induce between the two regions meeting there. Because $C^2 = 0$, the coherence operator δ_1 is the zero map, and the condition $\delta_1 r = 0$ holds automatically for every $r \in C^1$: with no triple overlaps, there is no local inconsistency a residue could exhibit, exactly as in the classical computation of Čech cohomology for a circular cover, where the obstruction lies entirely in whether a closed cochain is exact rather than in whether it is closed.

Consider the residue

$$r = (a_{12}, a_{23}, a_{34}, a_{41}) = (1, 1, 1, 1),$$

representing a configuration in which each pair of neighbouring regions reports a local mismatch of exactly one unit. Since δ_1 is the zero map, $\delta_1 r = 0$ holds trivially, so r passes the only coherence check available to this regional structure. We now show directly that $r \notin \operatorname{im} \delta_0$, so that no choice of local correction removes the mismatch. Suppose, for contradiction, that $\delta_0(c_1, c_2, c_3, c_4)^\top = r$ for some $c \in C^0$. Then

$$c_2 - c_1 = 1, \qquad c_3 - c_2 = 1, \qquad c_4 - c_3 = 1, \qquad c_1 - c_4 = 1.$$

Adding all four equations, the left-hand side telescopes to 0, since each c_i appears once with a positive and once with a negative sign, while the right-hand side sums to 4. This gives $0 = 4$, a contradiction. No correction c exists, and $r \notin \operatorname{im} \delta_0$. The class $[r] = r + \operatorname{im} \delta_0$ is therefore a non-zero element of $C^1 / \operatorname{im} \delta_0$, and the configuration carrying this residue is locally coherent at every overlap, in the sense that every pairwise comparison reports a consistent discrepancy and no triple-overlap check is available to flag a contradiction, while admitting no global assembly: there is no assignment of values to the four regions that simultaneously accounts for all four observed mismatches. This is the finite obstruction promised above, given as an explicit object rather than as a schema, and Appendix B records the same computation together with the general statement of which residues are obstructions and which are not for this regional structure.

This example is deliberately minimal. Because $C^2 = 0$, the coherence condition $\delta_1 r = 0$ holds automatically for every residue, so the obstruction here turns entirely on exactness, on whether r lies in $\operatorname{im} \delta_0$, rather than on any non-trivial interplay between δ_1 and δ_0. A richer cover, with regions overlapping in triples, would introduce a non-trivial δ_1 and a correspondingly richer obstruction theory, in which a residue could fail to be locally coherent at all. We do not need that richer case to make the point of this section. The one-dimensional obstruction exhibited here, classified completely in Appendix B, already shows that local coherence does not entail global coherence, which is the only claim this example needs to support.

5.5 The admissibility reading

The connection to Section 4 is now direct, provided the observation regime is specified correctly, and specifying it correctly is the discipline this paper recommends elsewhere. Relative to the weak local regime M_{loc}, two configurations x, y can be indistinguishable: both may present locally coherent profiles, with no contradiction visible to any check available inside that regime, while one carries an exact residue and the other carries a non-vanishing obstruction class. In that case

$$M_{\text{loc}}(x) = M_{\text{loc}}(y) \quad \text{but} \quad \Phi(x) \neq \Phi(y),$$

so the global-coherence predicate does not descend along M_{loc}, which is the inadmissibility condition of Definition 1 instantiated in a setting concrete enough to compute.

Relative to the richer certificate regime M_{cert}, by contrast, Φ is decidable: a verifier with access to δ_0 and r_x tests directly whether $r_x \in \operatorname{im} \delta_0$, and Proposition 3 guarantees that this test is equivalent to admissibility of Φ with respect to M_{cert}. The finite example therefore illustrates both halves of the paper's thesis at once, rather than only the negative half. A local, productive regime can be inadmissible for a predicate of genuine interest, and no amount of further local inspection repairs the defect, while a closure regime, built by supplying the obstruction data the local regime withheld and checking it independently, becomes adequate. The local observation regime records profile data sufficient to decide local coherence but insufficient, in general, to decide the global predicate; the predicate "admits a globally coherent assembly" fails to descend to the locally observed profile unless the obstruction class is computed, and computing it is what distinguishes M_{cert} from M_{loc}.

5.6 Warrant debt

We introduce here the central diagnostic quantity of the paper. Warrant debt is the residual obstruction left when local admissibility checks do not lift to global admissibility. In the finite case developed above, warrant debt is witnessed by the non-vanishing class $[r_x] \neq 0$ in the relevant cohomology of the regional system, and Appendix B makes this precise. Warrant debt is not a measure of how far a system is from finding a correct global assembly through further search. It is a certificate that no such assembly exists under the current regional data, obtained by a calculation independent of further search.

5.7 Certificate architecture

The finite example also previews the architectural proposal of Section 11 in miniature, and the two-regime structure of the preceding subsections is what that architecture requires. The residue r_x, together with δ_0, is the certificate: it is the data $M_{\text{cert}}(x) = (\delta_0, r_x)$ that a closure layer needs and that a productive system confined to M_{loc} does not by itself produce. A verifier independent of whatever process proposed the local assignments can check, by finite computation, whether this certificate vanishes. A productive system, in this picture, proposes regional configurations and reports what is visible under M_{loc}; a closure layer constructs or receives the certificate $M_{\text{cert}}(x)$ and audits the resulting warrant status by computing the residue and reporting whether it vanishes. We defer the general statement of this architecture to Section 11 and Appendix C, and note only here that the finite example already contains every element the general architecture requires. Section 11 also records an executable cyclic diagnostic instance of the same architecture, in which the certificate is emitted as data and independently checked rather than only described.

5.8 Why the finite example matters

The finite example does not prove that every artificial intelligence failure has this exact form. It is not offered as such a proof. It shows the precise kind of structural failure with which this paper is concerned: local success under a declared regime can coexist with global inadmissibility, and the difference between the two is detectable only by a closure operation that is not reducible to further local optimisation. The two sections that follow show that the same triple structure, a state space, an observation map, and a predicate that fails to descend, recurs in two of the most discussed failure modes in contemporary artificial intelligence. The observation maps used there, a reward trace in Section 6 and a benchmark score in Section 7, play the role of M_{loc} rather than M_{cert}: each records what a productive system has access to, not a certified obstruction witness. Constructing an artificial intelligence analogue of M_{cert}, an explicit, checkable closure certificate for reward hacking or benchmark failure, is the open problem Section 14 registers, and the finite case worked here is offered as a model of what such a certificate would need to supply.

6 Reward Hacking as Admissibility Failure

Reward hacking is the cleanest artificial intelligence instance of the structure developed above, because it already presents, in the existing literature, in the form of a proxy signal, an optimiser, an intended predicate, and a documented divergence between the two (Amodei et al., 2016;

Manheim and Garrabrant, 2018; Skalse et al., 2022). This section makes that structure explicit as an X, M, Φ triple.

6.1 The setting

Let X be the set of environment trajectories available to a reinforcement learning agent. A trajectory $x \in X$ includes the sequence of world states, the agent's actions, the environmental transitions induced by those actions, and any outcome facts relevant to the designer's intent, whether or not those facts are visible to the reward function. Let

$$M : X \to O$$

be the reward-observation map, where $M(x)$ returns the scalar reward trace accumulated along the trajectory together with whatever logged observations and proxy features were used in training. Let

$$\Phi : X \to \{0, 1\}$$

be the designer-intended success predicate, for instance $\Phi(x) = 1$ if and only if the trajectory accomplishes the intended task safely and without deception.

6.2 Admissibility and failure

The reward regime is admissible for the intended predicate only if

$$M(x) = M(y) \implies \Phi(x) = \Phi(y),$$

that is, only if any two trajectories with the same reward-observable profile agree with respect to intended success. The admissibility failure characteristic of reward hacking occurs when there exist trajectories x, y such that

$$M(x) = M(y) \quad \text{but} \quad \Phi(x) \neq \Phi(y).$$

Here x genuinely satisfies the intended task, whereas y receives an identical reward profile while violating it. By Proposition 1, no decision rule over O alone, and hence no optimiser whose objective is a function of O alone, can recover Φ when this occurs.

6.3 Interpretation

The ordinary description of this phenomenon says that the agent gamed the reward. The admissibility account gives a structurally prior diagnosis: the reward map was not an admissible observation regime for the intended predicate, and optimisation exposed the collapse rather than causing it. This is a genuine explanatory addition, not a relabelling, because it identifies the prior structural condition whose failure makes reward hacking possible in the first place, independently of any particular agent's strategy for exploiting it. The diagnosis is also independent of how the reward map is constructed. A reward model learned from human preference comparisons (Christiano et al., 2017) is still an observation map in the sense of Definition 1, and the admissibility obligation applies to it unchanged.

6.4 Correspondence with the finite obstruction

The correspondence with Section 5 is direct and is summarised in Table 1. Local coherence in the finite example corresponds to reward or proxy satisfaction along a trajectory. The global gluing predicate corresponds to designer-intended success. A non-zero residue corresponds to the divergence between proxy satisfaction and intention. The failure of further local correction corresponds to the failure of further reward optimisation to repair the divergence. The verifier that detects the obstruction in the finite case corresponds to a closure authority that detects inadmissibility in the reinforcement learning case, a correspondence developed architecturally in Section 11.

Finite obstruction	**Reward hacking**
Local coherence	Reward or proxy satisfaction
Global gluing predicate	Designer-intended success
Non-zero residue	Proxy or intention divergence
Local correction fails	Further reward optimisation fails
Verifier detects obstruction	Closure authority detects inadmissibility

Table 1: Correspondence between the finite obstruction example and reward hacking.

7 Benchmark and Proxy Failure as Admissibility Failure

This section gives the second worked artificial intelligence example. Benchmark and proxy failure is structurally closer to Goodhart's law, discussed in general terms in Section 8, and the worked triple given here prepares that discussion.

7.1 The setting

Let X be the set of model-task-world interactions under deployment, that is, the set of concrete encounters between a trained system and the situations it is asked to handle once deployed. Let

$$M : X \to O$$

be the benchmark observation map, where $M(x)$ records the benchmark score, test-set performance, or visible evaluation transcript associated with the system's behaviour on x. Let

$$\Phi : X \to \{0, 1\}$$

be the deployment adequacy predicate, for instance $\Phi(x) = 1$ if and only if the system is competent, safe, or reliable on the relevant deployment class to which x belongs.

7.2 Admissibility and failure

Benchmark adequacy requires

$$M(x) = M(y) \implies \Phi(x) = \Phi(y),$$

that is, systems or interactions indistinguishable by benchmark evaluation must not differ with respect to deployment adequacy. Benchmark or proxy failure occurs when there exist x, y with

$$M(x) = M(y) \quad \text{but} \quad \Phi(x) \neq \Phi(y),$$

meaning that two interactions look equivalent under the benchmark while one is deployment adequate and the other is not. No benchmark optimiser can guarantee Φ when Φ does not descend along M, by the same fixed-regime limit established in Section 4.

7.3 Correspondence with the finite obstruction

The correspondence with Section 5 holds here in the same shape as for reward hacking, and is summarised in Table 2. Local coherence in the finite example corresponds to agreement between systems on benchmark score. The global gluing predicate corresponds to deployment adequacy. A non-zero residue corresponds to the divergence between benchmark equivalence and deployment outcome. The failure of further local correction corresponds to the failure of further benchmark-targeted optimisation to repair this divergence. The verifier that detects the obstruction in the finite case corresponds, as in Section 6, to a closure authority that detects inadmissibility before deployment.

Finite obstruction	**Benchmark or proxy failure**
Local coherence	Agreement on benchmark score
Global gluing predicate	Deployment adequacy
Non-zero residue	Benchmark or deployment divergence
Local correction fails	Further benchmark optimisation fails
Verifier detects obstruction	Closure authority detects inadmissibility before deployment

Table 2: Correspondence between the finite obstruction example and benchmark or proxy failure.

7.4 Why benchmark-driven development can produce brittle systems

This explains a pattern widely noted in practice but rarely traced to its structural source. It is not only that a given benchmark is an imperfect proxy. The benchmark induces an observational quotient on the space of systems and interactions, and optimisation conducted over that quotient cannot recover distinctions the quotient removes, however much further optimisation is applied. Brittleness under distribution shift is, on this reading, the expected consequence of training against a regime that was never shown to be admissible for the deployment predicate it was meant to track.

7.5 A cautious extension to adversarial classification

The same structure offers, with caution, a partial reading of adversarial misclassification. Let X be the set of semantic image-world pairs, let $M(x)$ be the model-accessible representation of the input, and let $\Phi(x)$ be the semantic class predicate. A structural failure of the kind discussed here occurs if the representation used for classification identifies, or fails to separate, states that differ semantically. We offer this only as a partial reading. Adversarial robustness involves additional geometric and statistical structure that the admissibility framework does not by itself capture, and we do not present this paragraph as a complete theory of adversarial examples. It isolates one structural failure mode among several that the adversarial robustness literature has identified, namely that semantic distinctions required by Φ may fail to be preserved by the operative representation M.

8 Goodhart's Law as Dynamic Inadmissibility

The two worked examples of Sections 6 and 7 now permit a structural reading of Goodhart's law that goes beyond the behavioural description standard in the literature (Goodhart, 1984). Discussions of this principle in machine learning typically interpret it behaviourally, in terms of agents that game metrics, overfit benchmarks, or exploit weaknesses in evaluation procedures (Manheim and Garrabrant, 2018). These descriptions are accurate as far as they go. They describe symptoms, and they do not by themselves identify the structural condition under which the symptoms become inevitable.

A measure is an observation map in the sense of Definition 1. Let $M : X \to O$ assign to each state a measurable output, a score, a metric, or a signal, of the kind that benchmark accuracy, standardised test performance, and economic performance indicators all instantiate. A target is the maximisation, or minimisation, of a function on O rather than directly on X. The implicit assumption underlying this procedure is that M preserves the distinctions in X relevant to the underlying objective Φ. Goodhart's law is the empirical observation that this assumption frequently fails, and the account given here identifies why the failure is not contingent.

Recall the equivalence relation $x \sim_M y \iff M(x) = M(y)$ from Section 4. Any optimisation process operating solely on M can move only within or between the blocks of this relation. It cannot recover a distinction M has already deleted. Where Φ is the property of interest, M is adequate to Φ only when Φ is constant on the blocks of $\sim_M$. Where this fails, Φ does not factor through M, and Proposition 1 applies directly.

In the absence of optimisation pressure the mismatch between M and Φ may remain latent, because most states lie far from the extremal regions of O and the equivalence classes there may happen to be relatively homogeneous with respect to Φ. Once M becomes a target, the system is driven toward the extremal regions of O, where the equivalence classes of $\sim_M$ frequently contain states that differ sharply with respect to Φ. Optimisation accordingly produces systematic drift toward states that maximise $M(x)$ without regard to whether $\Phi(x)$ holds. This is not an accidental failure and not a consequence of insufficient data. It is the structural consequence of M's failure to preserve the distinctions Φ requires, a consequence made visible once optimisation pushes the system into the region where that inadequacy is most pronounced.

Goodhart's law admits, on this reading, a sharper formulation. When a measure becomes a target, optimisation exposes the inadmissibility of the underlying predicate with respect to the observation map. Reward hacking, as set out in Section 6, is the case in which M is a reward trace. Benchmark gaming, as set out in Section 7, is the case in which M is a benchmark score. Proxy failure more generally is the case in which M is any proxy metric standing in for an intended predicate. In each case optimisation pushes the system through the quotient induced by M until the lost distinctions matter, and the resulting drift is what an external observer records as Goodhart's law.

Goodhart's law is what admissibility failure looks like when a proxy is made operational and optimisation is allowed to run against it.

9 Boundary Revision as Philosophical Motivation

The formal results above do not establish an empirical theory of human randomness, and this section does not claim to supply one. Its purpose is narrower: to show why intelligence in the stronger sense relevant to questions of artificial general capability may require operations on observational regimes, not only optimisation within them, and to use this possibility as philosophical motivation for the architectural proposal of Section 11.

9.1 Apparent randomness under non-injective observation

The following proposition is offered as a modelling caution rather than as a piece of cognitive science. Let $M : X \to O$ be an observation map and let $f : X \to Y$ describe some target behaviour. If there exist x, y with $M(x) = M(y)$ but $f(x) \neq f(y)$, then no $g : O \to Y$ satisfies $f = g \circ M$. From the standpoint of an observer restricted to O, the behaviour governed by f therefore appears non-functional, and hence stochastic-looking, even where f itself is perfectly determinate on X. The proof uses the same fibre argument as Proposition 1 and is recorded as Proposition 4 in Appendix A for completeness.

9.2 An interpretive point, stated carefully

We do not claim that human randomness is boundary revision. The weaker claim is enough: behaviour classified as random relative to one regime may reflect competence under distinctions that regime does not preserve. Collapsing the two claims would overstate the formal result.

9.3 A criterion against unfalsifiability

A hypothesis of this kind is useful only if it can be distinguished from noise. The criterion is this: a boundary-revision interpretation is strengthened when behaviour unpredictable under the old regime becomes stable, intelligible, and repeatably competent under a later description. Noise remains unstable under refinement. The test is not whether the behaviour first appears surprising, but whether a later description makes it coherent.

Definition 2 (Admissible futures relative to an observation map)**.** *Let X be a state space, let $M : X \to O$ be an observation map, let $\Phi : X \to \{0, 1\}$ be the intended predicate, and let $\mathcal{F}$ be a class of candidate futures, continuations, or interventions $F : X \to X$. For $s \in X$, define*

$$\mathrm{AFut}_{M,\Phi}(s) = \{\, F \in \mathcal{F} \mid \forall x \in X,\ M(x) = M(s) \Rightarrow \Phi(F(x)) = \Phi(F(s)) \,\}.$$

Thus a future is admissible relative to M at s only when the resulting Φ-judgement is well defined over the whole observational fibre of s.

Theorem 1 (Refinement monotonicity of admissible futures)**.** *Let $M : X \to O$ and $M' : X \to O'$ be observation maps. Suppose that M' refines M, in the sense that there exists a comparison map $\rho : O' \to O$ such that*

$$M = \rho \circ M'.$$

Then, for every $s \in X$,

$$\mathrm{AFut}_{M,\Phi}(s) \subseteq \mathrm{AFut}_{M',\Phi}(s).$$

Moreover, the inclusion is strict only when the refinement separates an ambiguity that was collapsed by M and that is relevant to the Φ-judgement of some continuation in $\mathcal{F}$.

Proof. Since $M = \rho \circ M'$, any equality $M'(x) = M'(s)$ implies

$$M(x) = \rho(M'(x)) = \rho(M'(s)) = M(s).$$

Hence the M'-fibre of s is contained in the M-fibre of s. Any continuation that preserves the required Φ-distinction uniformly over the larger M-fibre therefore preserves it uniformly over the smaller M'-fibre. This proves

$$\mathrm{AFut}_{M,\Phi}(s) \subseteq \mathrm{AFut}_{M',\Phi}(s).$$

Strict inclusion requires more than refinement alone. It occurs only when some continuation is inadmissible under the coarser map M because M identifies states whose images under that continuation receive different Φ-values, while the refined map M' separates those states. □

This gives a falsifiable criterion for boundary revision. A proposed revision is not vindicated merely because behaviour appears surprising under the old regime. It must exhibit a refined observation map, a comparison back to the old map, and an enlarged class of admissible futures. Noise need not satisfy this condition. A genuine structural-revision claim must be able to supply it.

Boundary revision can be given a structural reading without adding a topological calculus of regime change to this paper. Optimisation pressure may expose a mismatch between the region selected by a proxy and the region on which the intended predicate is stable. In the terms used here, the failure is that the proxy-induced partition no longer preserves the distinction required by the target predicate. A mereotopological treatment would require separate definitions of region, interior, closure, and cover; those definitions are left for another paper.

9.4 Illustrations

The following cases are illustrations, not evidence for an empirical thesis. Scientific revolutions in the sense described by Kuhn appear, from inside the abandoned paradigm, as a refusal to play by the old rules (Kuhn, 1962). Mathematical concept revision of the kind documented by Lakatos, in which the concept of a polyhedron is not held stable across the proof-and-refutation history of Euler's formula, shows a concept being revised by the very proof that was meant to be a proof about it (Lakatos, 1976). Strategic reframing in competitive settings, moral refusal against an aggregated prior calculus of permitted reasons, and creative reframing more generally all share the form: behaviour that appears incoherent under the prior frame becomes coherent under a newly articulated one. Peirce's account of abduction as a third inferential mode, irreducible to deduction or induction, and Whitehead's account of creative advance, are the philosophical traditions from which this reading is drawn (Peirce, 1903; Whitehead, 1929).

9.5 Relevance to artificial intelligence

The relevance to artificial intelligence is architectural, not imitative. The paper does not claim that machines must imitate human spontaneity. Systems operating in consequential settings require some mechanism for detecting when their current regime has ceased to preserve the relevant distinctions, whatever the ultimate explanation of human cognitive flexibility turns out to be. Section 10 develops the structural contrast this motivates, and Section 11 develops the architecture that answers it.

10 Artificial and Natural Intelligence: A Structural Contrast

10.1 The contrast is architectural

The contrast at issue in this section is not between carbon and silicon, nor between consciousness and mechanism. It is between architectures restricted to productive operations within a regime and architectures that include operations on regimes, in the sense developed in Section 3.

10.2 Contemporary artificial systems

Contemporary artificial intelligence systems can learn representations, adapt parameters, and alter internal encodings over the course of training. That point is not contested. But representational change produced by optimisation within a fixed training objective is not certified repair of inadmissibility. A model that updates its weights in response to gradient signal is still, by the construction of that update rule, a productive system operating within whatever regime the training objective and architecture jointly define. Updating within a regime and revising the regime are different obligations. The first does not discharge the second.

10.3 Natural intelligence as hypothesis

Natural intelligence appears, in some cases, to include regime-revising capacities of the kind discussed in Section 9. We state this as a structural hypothesis, not as an empirical result established by this paper. The hypothesis is offered to give the natural-intelligence side of the contrast a precise formal shape against which architectural alternatives, artificial or otherwise, can be compared, and not as a settled finding about human cognition.

10.4 Plural observational structure

Suppose an agent's representational structure consists not of a single observation map but of a family $\{M_i : X \to O_i\}_{i \in I}$ of partially independent observation maps, each preserving a different set of distinctions and each admissible, by hypothesis, for predicates appropriate to its own scope. A coherence strain occurs when two or more maps in the family make incompatible determinations of a predicate that lies within their shared scope, even though each map is individually admissible for its own restricted purpose. A coherence strain is not the failure of any single map. It is a structural signal that local adequacy, here adequacy of each M_i taken alone, does not lift to joint adequacy across the family, a phenomenon directly analogous to the finite obstruction of Section 5, where local coherence likewise failed to lift to global coherence.

10.5 Why plural structure matters

A single observation map may collapse distinctions that another map in the family preserves. Behaviour governed by the second map will then appear random or irrational from the standpoint of the first, by Proposition 4. This gives a structural account of why the same behaviour can appear principled under one description and arbitrary under another, without requiring that either description be mistaken about the facts it has access to.

10.6 A statement against overclaim

None of this proves that human cognition has the plural structure sketched above. The hypothesis gives a formal shape to the kind of structure that would be required for boundary revision to operate as Section 9 describes. Whether human cognition, or any other candidate instance of natural intelligence, in fact realises this structure is left open, and the architectural proposal of the next section does not depend on the hypothesis being correct.

11 Closure Architecture: Warrant Debt, Certificates, and Independent Verification

11.1 Productive layer and closure layer

We now state the architecture motivated by Sections 3 through 9. The proposal separates three functions, set out in Table 3: a productive layer that proposes, predicts, acts, and optimises; a closure layer that checks admissibility and detects warrant debt; and a reconstruction authority that revises or escalates the observational regime when the closure layer finds it inadequate.

Layer	Function
Productive layer	Proposes, predicts, acts, optimises
Closure layer	Checks admissibility, detects warrant debt, rejects inadequate regimes
Reconstruction authority	Revises or escalates the observational regime

Table 3: The three architectural layers of a closure-disciplined system.

11.2 Warrant debt, restated generally

Warrant debt, introduced in Section 5, is the residual gap between local or proxy satisfaction and warranted admissibility. In finite settings of the kind given in Section 5 and Appendix B, warrant debt is represented by a non-vanishing residue class $[r] \neq 0$. In broader artificial intelligence settings of the kind discussed in Sections 6 and 7, an exact computable residue of this kind is not yet available, but the concept still identifies the target that any detection method must aim at: a quantity, however represented, that certifies the absence rather than only the as-yet-unfound presence of a globally adequate assembly.

11.3 Monitorability as admissibility

The closure architecture can also be stated in monitoring language. A predicate $\Phi : X \to \{0,1\}$ is monitorable from the observation map $M : X \to O$ when there exists a monitor

$$J : O \to \{0,1\}$$

such that

$$\Phi = J \circ M.$$

Thus monitorability is not an additional condition beyond admissibility. It is admissibility read operationally: the observation stream contains enough information for the monitor to decide the predicate.

Consequently, non-monitorability is not a failure of the monitor's ingenuity. It is a structural property of the observation map. If there exist $x, y \in X$ with

$$M(x) = M(y) \quad \text{but} \quad \Phi(x) \neq \Phi(y),$$

then no monitor operating on O alone can decide Φ. This is the monitoring form of Proposition 1, and Proposition 2 adds that no downstream post-processing of the same observation stream can change that verdict. An inadequacy signal, in the architectural sense used below, is therefore evidence that the intended monitor is being asked to decide a predicate that does not descend along the observation regime it receives.

11.4 Inadequacy signals

Definition 3 (Structural divergence)**.** *Let $M : X \to O$ be the operative observation map and let $\Phi : X \to \{0,1\}$ be the intended predicate. A structural divergence witness for (M, Φ) is evidence of two states $x, y \in X$ such that*

$$M(x) = M(y) \qquad \textit{but} \qquad \Phi(x) \neq \Phi(y).$$

Equivalently, it is evidence that the current observational fibre contains states that require different judgements under the intended predicate. A structural divergence witness is thus precisely a witness that admissibility in the sense of Definition 1 fails for the pair (M, Φ).

Structural divergence is the formal shape of an inadequacy signal. It is not merely a warning that the system has made an error. It is evidence that the system is being asked to decide a predicate that does not descend along the observation regime it receives. In settings where Φ is not directly observable, a structural divergence witness may be indirect: proxy success paired with intended failure, of the kind made explicit in Section 6, persistent divergence under distribution shift, benchmark equivalence paired with deployment divergence, of the kind made explicit in Section 7, local consistency paired with a non-vanishing global obstruction, of the kind exhibited in Section 5, or instability within a single observational fibre that does not resolve under further training.

In the declared finite setting of this paper, structural divergence has a certificate-level form. A coherent local residue r satisfying

$$\delta_1 r = 0$$

but failing the exactness condition

$$r \notin \operatorname{im} \delta_0$$

is a finite witness that local compatibility does not lift to global warrant. Equivalently, let D denote the squared norm of the orthogonal projection of r onto the harmonic subspace

$\ker \delta_1 \cap (\operatorname{im} \delta_0)^\perp$. Since $\operatorname{im} \delta_0 \subseteq \ker \delta_1$, a coherent residue decomposes orthogonally into an exact part and a harmonic part, so for coherent residues a positive harmonic debt magnitude

$$D > 0$$

holds precisely when $r \notin \operatorname{im} \delta_0$, and this positive value is the exported finite inadequacy signal. In the one-dimensional obstruction setting of the prototype below, D reduces to $\langle z, r\rangle^2/\|z\|^2$ for the harmonic representative z. It gives the verdict WARRANT-DEBT. By contrast, a coherent residue with zero harmonic period gives GLOBALLY-ADMISSIBLE, while a failure of the cocycle condition gives the separate verdict COHERENCE-FAILURE.

We emphasise, to avoid suggesting more completeness than the paper delivers, that outside declared finite settings these are classes of signals indicating where a detection algorithm must operate, not a complete neural detection algorithm in themselves. The unresolved problem is not whether structural divergence can be stated. It can. The unresolved problem is how to extract reliable structural divergence witnesses from systems whose effective observation maps are implicit in learned weights rather than declared in the architecture.

11.5 Finite warrant-debt detection

The finite case admits a complete algorithm. Stating it marks the boundary between what this paper supplies and what remains open for learned systems whose observation maps are implicit. In the declared finite setting, the closure layer is not performing an open-ended search for better behaviour. It is deciding a precise membership question:

$$r \in \operatorname{im} \delta_0?$$

where $\delta_0 : C^0 \to C^1$ is the correction operator and $r \in C^1$ is the residue associated with the candidate configuration.

Algorithm 1: Finite Warrant-Debt Detection

Input. A finite-dimensional correction operator $\delta_0 : C^0 \to C^1$, a residue $r \in C^1$, and, where the regional structure has non-trivial higher overlaps, a coherence operator $\delta_1 : C^1 \to C^2$.

Output. One of three verdicts:

COHERENCE-FAILURE, GLOBALLY-ADMISSIBLE, WARRANT-DEBT.

1. Check that the dimensions of δ_0, r, and, where present, δ_1 agree.
2. If δ_1 is present and $\delta_1 r \neq 0$, return

   COHERENCE-FAILURE.

   The reported residue is not locally coherent.
3. Solve the exact linear system
   $$\delta_0 c = r.$$
4. If a solution c exists, return

   GLOBALLY-ADMISSIBLE

   with c as a repair witness.

5. If no solution exists, compute a non-zero left-nullspace witness h satisfying

$$h^\top \delta_0 = 0$$

and

$$h^\top r \neq 0.$$

6. Return

$$\textsc{Warrant-Debt}$$

with h as an obstruction witness.

Algorithm 1 is sound and complete for finite-dimensional linear obstruction systems over an exact field such as $\mathbb{Q}$. Soundness is immediate. If the algorithm returns a correction c, then $r = \delta_0 c$, so the residue is exact. If it returns a left-nullspace witness h, then any exact residue $\delta_0 c$ satisfies $h^\top \delta_0 c = 0$, while the reported residue satisfies $h^\top r \neq 0$; hence $r \notin \operatorname{im} \delta_0$. Completeness follows from finite-dimensional linear algebra: if $r \notin \operatorname{im} \delta_0$, then there is a linear functional separating r from $\operatorname{im} \delta_0$, equivalently a vector $h \in \ker(\delta_0^\top)$ with $h^\top r \neq 0$ under the standard pairing.

For the four-cycle example of Section 5 and Appendix B, the witness is simply

$$h = (1, 1, 1, 1),$$

so that $h^\top \delta_0 = 0$ while $h^\top r = 4 \neq 0$ for $r = (1, 1, 1, 1)$. The verdict is therefore not a heuristic judgement and not a timeout in a search procedure. It is a finite certificate that no local correction exists.

11.6 A finite executable prototype

The finite example of Section 5 instantiates the proposed architecture. A productive system generates or reports a candidate state. A closure auditor extracts a certificate from that state. The auditor checks, by finite computation where the setting is finite, whether the certificate is globally admissible. The result is not a confidence score. It is one of three accountable verdicts:

$$\textsc{Coherence-Failure}, \qquad \textsc{Globally-Admissible}, \qquad \textsc{Warrant-Debt}.$$

An independent verifier, separate from the auditor that produced the original finding, confirms the result before it is acted upon.

The accompanying finite prototype implements this discipline for a cyclic diagnostic system. The setting is a four-interface cyber-physical diagnostic loop, with a sensor, controller, actuator, and monitor joined at those interfaces. The productive diagnostic layer proposes the claim

```
fault_origin = B_Controller.
```

The closure layer does not accept or reject that claim by extending the same search. It constructs the cyclic residue, computes its harmonic period, and emits a certificate. In this executable instance the first three coordinates follow the forward diagnostic interfaces sensor-to-controller, controller-to-actuator, and actuator-to-monitor, while the closing interface is recorded in the opposite geometric direction, sensor-to-monitor rather than monitor-to-sensor. Thus the signed edge basis differs from the abstract four-cycle convention of Appendix B: the final row of the correction matrix is $(-1, 0, 0, 1)$, and in this basis the harmonic cycle representative is $z = (-1, -1, -1, 1)$.

The emitted fault residue is

$$r = (1, 1, 1, -2).$$

The harmonic period is therefore

$$\langle z, r\rangle = (-1)(1) + (-1)(1) + (-1)(1) + (1)(-2) = -5,$$

and the warrant-debt magnitude is

$$D = \frac{\langle z, r\rangle^2}{\|z\|^2} = \frac{25}{4}.$$

The vector r records the emitted discrepancy on these four signed interfaces, and D is the warrant-debt magnitude, namely the squared length of the projection of r onto the harmonic direction. The positive value $D > 0$ is the finite inadequacy signal. The verdict is therefore

WARRANT-DEBT.

The local diagnostic residues are coherent, but the global attribution claim is not warranted by the declared observation regime. This is the finite operational counterpart of the admissibility failure described abstractly above: the defect is not a failure to search longer for a better attribution, but a non-zero obstruction witnessed by the certificate.

The same prototype also contains a refinement case. After an additional sensor distinction is supplied, the residue changes so that the cyclic period vanishes:

$$\langle z, r\rangle = 0, \qquad D = 0.$$

The corresponding verdict is

GLOBALLY-ADMISSIBLE.

This gives a finite example of regime revision in the modest sense required by the present paper. The observation regime is not made more successful by a cleverer downstream classifier operating on the same collapsed data. It is refined so that the obstruction detected by the closure layer vanishes.

A third case in the prototype distinguishes warrant debt from ordinary local inconsistency. In that case a triple-overlap check gives

$$\delta_1 r \neq 0,$$

and the verdict is

COHERENCE-FAILURE.

The H^1 obstruction question does not arise because the local reports are already mutually inconsistent. This separation matters architecturally. A closure layer should not collapse all failure into one undifferentiated error state. It must distinguish incoherent evidence, coherent but globally unwarranted evidence, and evidence whose obstruction has vanished.

In implementation terms, the finite prototype is a certificate-verifier pipeline. The front end constructs cyclic diagnostic examples and emits JSON certificates. The certificate contains the declared incidence structure, the residue vector, the harmonic basis data, the stored period, the stored debt magnitude, the cocycle flag, and the claimed verdict. A separately implemented verifier recomputes the finite linear-algebra check from the certificate alone. In the current implementation the independent path is realised by an exact OCaml/Zarith verifier, together with a Rocq certificate core and a Rocq-extracted verifier for the parts of the check already mechanised. The important architectural property is not the programming language used. It is that the productive diagnostic process, the emitted certificate, and the closure verdict are separated. The verifier is not asked to trust the process that proposed the diagnostic claim.

The supporting materials also include a broader Rocq/OCaml formalisation of the structural admissibility layer. This companion development mechanises the factorisation criterion, safety-relevant collapse, warrant-debt witnesses, refinement and restriction repairs, runtime rupture certificates, and small declared-regime case studies in physical sensing, distributed consensus, and GPU memory consistency. Its role is different from the finite CycleDebt prototype. The prototype supplies the concrete numerical certificate used in this section; the companion formalisation supplies a reusable proof-oriented account of the admissibility and warrant-debt concepts that the prototype instantiates.

This executable prototype answers a limited but important version of the methodological objection. It supplies a concrete finite method for detecting inadmissibility in a declared regime, a numerical inadequacy signal $D > 0$, a certificate format, three operational verdicts, and an example in which refinement of the observation regime removes the obstruction. It does not solve the harder neural problem. The property to generalise is this: the proposal, the certificate, and the verdict are all checkable objects. Section 14 records why this property is still open for systems whose observation maps are not declared.

11.7 What a successful implicit-map algorithm would have to supply

For systems whose observation map is implicit in learned weights, hidden activations, attention patterns, reward channels, or benchmark interfaces, a complete algorithm cannot begin with a declared $M : X \to O$ in the way Algorithm 1 does. The algorithmic question is then different, not meaningless. A successful algorithm for implicit learned regimes would have to turn the hidden or distributed observational regime into something that can be audited.

The general pattern is not to make the monitor cleverer while the observation remains fixed. The pattern is that the observation itself must be refined. In the finite example this is the passage from M_{loc} to M_{cert}. In a learned system the corresponding problem is to construct an exposed or auditable observation regime $\widehat{M}$ that records the distinctions, witnesses, or invariants needed to decide the target predicate. A successful implicit-map algorithm would not only classify outputs. It would either produce such a refined observation regime, together with a checkable admissibility witness, or return No-Certificate.

One possible route is to construct auditable surrogate observation maps from learned representations. For neural systems, the operative regime is rarely given as a declared map $M : X \to O$. It is distributed across embeddings, hidden activations, attention patterns, loss functions, reward channels, and output interfaces. A successful surrogate method would have to extract from these distributed structures a lower-dimensional or symbolic map $\widehat{M}$ whose fibres can be probed for admissibility-relevant variation. Topological data analysis offers one possible family of tools for this task. By studying the persistent structure of activation spaces under input perturbation, one may attempt to identify which distinctions are preserved by the learned representation and which are collapsed. Such a surrogate would not prove admissibility by itself. Its role would be to make the effective observational quotient inspectable enough that contrastive witnesses, fibre instability, or certificate failures can be searched for. In the terminology of this paper, the aim is not interpretation alone, in the sense pursued by the interpretable machine learning literature (Doshi-Velez and Kim, 2017). The aim is to expose an auditable candidate $\widehat{M}$ on which admissibility can be tested (Edelsbrunner et al., 2002; Carlsson, 2009).

Such an algorithm would need, at minimum, the following components.

1. **Regime exposure.** It must identify, extract, or impose an auditable surrogate for the operative observation map. This surrogate need not reproduce every internal parameter of the model, but it must specify which distinctions the system treats as preserved and which it treats as collapsed for the task under inspection.

2. **Predicate specification.** It must receive a target predicate Φ, or a decision-relevant surrogate for Φ, from outside the productive component. The productive system cannot be the sole authority on the predicate whose admissibility is being checked.

3. **Fibre or equivalence probing.** It must characterise, generate, or approximate states that the exposed regime treats as equivalent. In the deterministic case this means searching for pairs x, y with $M(x) = M(y)$; in stochastic or learned settings it means searching for states that induce the same relevant observational law or are indistinguishable under the declared surrogate.

4. **Contrastive witness search.** It must attempt to find a witness of inadmissibility:

$$M(x) = M(y) \quad \text{while} \quad \Phi(x) \neq \Phi(y),$$

or the stochastic analogue in which two states are observationally equivalent for the model but differ with respect to the decision-relevant distribution or predicate.

5. **Positive certification.** If no counterexample is found, the algorithm must not simply report success. It must produce a positive certificate showing that, over a declared domain and relative to a declared surrogate regime, the target predicate is invariant over the relevant observational fibres. Without such a certificate, the correct verdict is not admissibility but absence of certification.

6. **Independent verification.** The certificate, whether positive or negative, must be checkable by a process independent of the productive component. A confidence score emitted by the model is not an admissibility witness, because it is itself a function of the regime being audited.

7. **Escalation discipline.** The algorithm must distinguish at least three outcomes:

$$\textsc{Certified-Admissible}, \qquad \textsc{Warrant-Debt}, \qquad \textsc{No-Certificate}.$$

The third verdict is essential. In consequential settings, failure to produce an admissibility certificate must not be silently converted into permission to proceed.

This subsection does not claim to supply a complete neural inadmissibility detector. It specifies what success would require. The finite algorithm above shows what completeness looks like when the regime is declared and the obstruction problem is linear. The implicit-map case remains open precisely because the first step, exposing a regime that can be checked without trusting the productive system's own self-report, is not solved in general.

11.8 The proof-assistant connection, restated

The proof-assistant analogy introduced in Section 3 can now be stated without overreach. The analogy is not autonomous foundation-revision. It is separation of proposal from certification: tactics correspond to productive search, the kernel to closure checking, kernel reconstruction to external regime revision, and independent verification to auditability.

The same separation appears in modern formal verification (Leroy, 2009). In the Rocq proof assistant, a tactic may search heuristically through a search space of possible derivations and propose a proof term. The tactic, however, has no authority to admit the result into the formal environment. That authority is reserved for a small kernel, which checks the proposed term against the rules of the underlying calculus (Coquand and Huet, 1988; Paulin-Mohring, 1993). The kernel does not search; it certifies. The closure layer follows the same principle. A productive component may generate, optimise, or propose; the closure component is restricted to checking whether the proposed state is admissible under the declared regime, or whether a warrant-debt certificate forces rejection or escalation.

The executable prototype uses the same separation in a finite diagnostic setting. The Python diagnostic engine proposes and records candidate cyclic states. The certificate is a portable object. The independent verifier checks algebraic conditions over exact rational data. The Rocq core mechanises the soundness of the verdict logic for the currently formalised checks, while the remaining linear-algebraic reconstruction of the harmonic direction is checked externally by the independent verifier. This qualification is important. The point claimed here is not that the implementation is a finished universal kernel for artificial intelligence systems, but that the architecture can be made operational in a finite declared regime without relying on the productive component's self-report.

11.9 What this does and does not answer

The paper now gives a complete algorithm only for finite declared obstruction systems, where the relevant spaces, operators, and residues are available as checkable objects. It also gives an executable cyclic diagnostic prototype of that finite algorithm. It does not provide a complete algorithm for learned neural systems whose observation maps are implicit in distributed weights rather than declared in their architecture. What it does identify is the required architectural form and the success conditions such an algorithm would have to satisfy. Systems intended for consequential use must produce inspectable admissibility witnesses, of the kind exhibited concretely for the finite case in Section 5, Algorithm 1, and the finite prototype above, rather than confidence scores alone, which by construction record only a function of the regime already in force and cannot certify the adequacy of that regime. We return to this distinction, and to what remains open about closing it, in Section 14.

12 Related Work: Abstraction, Monitoring, Partial Observability, and Proxy Failure

The admissibility framework intersects with several independently established research programmes, and this section situates the contribution accordingly.

The oldest precursor is cybernetic. Ashby's law of requisite variety states that a regulator can succeed only when its repertoire of distinctions matches the variety of the disturbances it must counter (Ashby, 1956). Admissibility is the descent-theoretic form of a related obligation: an observation channel whose induced partition lacks the variety to separate Φ-relevant states cannot support judgement of Φ, however capable the regulator operating downstream of the channel. In the philosophy of information, situation semantics likewise treats what an agent can discriminate or convey as relative to the informational situation the agent occupies, rather than as an intrinsic property of the agent's reasoning capacity (Barwise and Perry, 1983). The present framework shares that relativisation while restricting itself to the single formal question of whether a declared predicate descends along a declared observation map.

Abstract interpretation, originating with Cousot and Cousot, studies the relation between a concrete semantics and an abstraction map that necessarily collapses distinctions present in the concrete domain (Cousot and Cousot, 1977). The admissibility question, which properties survive passage through an abstraction or observation map, generalises this concern beyond the analysis of programs to the deployment of learning systems, while the underlying obligation, descent of a property along the partition the abstraction induces, is the same in both settings.

Work on runtime verification and monitorability asks which properties of a system can be decided from observations alone. The admissibility result of Section 4 can be read as a foundational statement of the same obligation: a property whose truth depends on distinctions an observation channel has collapsed is, in the sense of Proposition 1, not monitorable through that channel.

Partially observable Markov decision processes formalise decision-making when an agent does not observe the full state of its environment (Kaelbling et al., 1998). The present framework isolates a prior logical condition that POMDP theory itself does not by itself resolve: whether the observation available to the agent preserves the distinctions required by the predicate of interest, independently of how the resulting partial-information decision problem is then solved.

Representation learning can change the effective observation map over the course of training, refining or transforming M as parameters update (Bengio et al., 2013). The admissibility question survives this refinement unchanged in form: does the learned representation, whatever its provenance, preserve the distinctions required by the target predicate? Optimisation-driven representational change does not, by itself, certify that the resulting representation is admissible, a distinction often left implicit in representation-learning approaches to robustness.

The literature on proxy failure and specification gaming is the closest existing body of work to the present proposal, and Sections 6 through 8 draw directly on it, including the broad catalogue of empirically observed specification-gaming behaviour assembled by Krakovna et al. (Krakovna et al., 2020). The contribution here is to frame proxy failure as the case in which a proxy observation map fails to preserve an intended predicate, and to supply the finite worked obstruction of Section 5 as a concrete, checkable instance of the general phenomenon that literature has documented empirically. The analysis of learned optimisation identifies a related structural gap between a base objective and the objective implicitly pursued by an optimiser that training produces (Hubinger et al., 2019); in the present terms, the concern is that the effective observation regime and predicate of the inner optimiser need not coincide with those declared by the training procedure, which is a second place where an operative regime can diverge from a declared one. Broader treatments of the alignment problem (Russell, 2019; Ngo et al., 2022) emphasise the difficulty of specifying objectives that capture intended behaviour; the present paper isolates one formal component of that difficulty, namely that a specified objective is decidable only relative to an observation regime that is admissible for it.

Causal identifiability, in the tradition associated with Pearl, is structurally analogous to admissibility but is not reducible to it: a causal query is identifiable only when the available observational regime contains sufficient information to determine the quantity of interest, which is the structural form of admissibility transferred to the causal setting (Pearl, 2009). We do not claim that the do-calculus is a calculus of admissibility transfers in any technical sense beyond this structural analogy, and we leave a precise comparison of the two formalisms to future work.

Finally, in stochastic settings the analogue of admissibility is not literal equality on fibres but sufficiency: the observation must retain all information relevant to the predicate or decision problem at hand, in the sense familiar from the classical theory of sufficient statistics and its formulation in terms of the comparison of statistical experiments (Fisher, 1922; Blackwell, 1953). Section 13 develops this analogy as a sketch of the generalisation the deterministic theory of Section 4 admits.

13 Stochastic Observation and Sufficiency

The formal core of Section 4 uses deterministic observation maps. Most artificial intelligence systems involve stochastic observations, partial observability, and learned latent representations, so the deterministic case must be read as the base case, not the full theory. This section sketches the generalisation without claiming to complete it. The passage from deterministic observation maps to stochastic observation kernels is the passage from exact factorisation to probabilistic sufficiency. In the deterministic setting, admissibility requires that the target predicate be constant on the fibres of $M : X \to O$. Equivalently, the predicate factors through the observational quotient. In the stochastic setting, the corresponding requirement is that the observation retain all information relevant to the predicate or decision problem.

Let $K(o \mid x)$ be a stochastic observation kernel, replacing the deterministic map $M : X \to O$ with a conditional distribution $K : X \rightsquigarrow O$. The direct stochastic analogue of Definition 1 replaces equality of observations with equality of observational laws: a predicate $\Phi : X \to \{0, 1\}$

is exactly admissible with respect to K when

$$K(\cdot \mid x) = K(\cdot \mid y) \implies \Phi(x) = \Phi(y)$$

for all $x, y \in X$, that is, when no two states inducing the same distribution over observations differ in Φ. This is the literal stochastic counterpart of Definition 1, with the deterministic equality $M(x) = M(y)$ replaced by equality of the induced laws $K(\cdot \mid x) = K(\cdot \mid y)$, and it recovers Definition 1 when $K(\cdot \mid x)$ is a point mass at $M(x)$ for some deterministic M.

This exact condition is often stronger than what a decision problem actually requires, since two states can induce different observational laws while nonetheless agreeing on the decision-relevant distribution of the target. A weaker, decision-theoretic version asks that the conditional law of the target given the full state be recoverable from the observation. In this sense, stochastic admissibility is a sufficiency condition: the observation must not integrate out distinctions on which the predicate depends. Where it does, no downstream probabilistic calibration can restore the lost warrant; it can only quantify uncertainty inside an inadequate observational regime. Both the exact and the relaxed conditions reduce to Definition 1 in the deterministic limit, and the deterministic admissibility theorem of Section 4 is accordingly a limiting case of this broader sufficiency problem rather than a special case selected for convenience.

A full stochastic theory, with its own factorisation theorem and its own analogue of the fixed-regime limit, is beyond the scope of this paper. The purpose of this section is narrower: to show that the deterministic theorem of Section 4 is not a dead end restricted to an idealised case, but the limiting instance of a more general sufficiency problem that a complete treatment of admissibility for learned, stochastic, and partially observable systems will eventually have to address directly.

14 Limitations and Open Problems

We state plainly what the paper establishes and what it does not, in order to avoid claiming a completeness the argument does not possess.

The paper establishes a formal admissibility condition together with the elementary but consequential proposition that inadmissibility cannot be optimised away. It establishes a finite obstruction example, worked through explicitly in Section 5 and Appendix B, in which local adequacy provably fails to lift to global warrant. It establishes an executable finite prototype for a cyclic diagnostic system, in which local incoherence, global admissibility, and warrant debt are distinguished by certificate checks. It establishes two artificial intelligence schemata, reward hacking and benchmark or proxy failure, each written as an explicit X, M, Φ triple and each shown to instantiate the same obstruction structure as the finite case. It establishes an architectural discipline, set out in Section 11, for separating productive optimisation from closure authority. It establishes, finally, a research programme organised around the production of admissibility witnesses, rather than a finished solution to the problem the programme addresses.

The paper does not yet provide a general algorithm for detecting inadmissibility in neural networks whose observation maps are implicit in distributed weights rather than declared in their architecture. The executable prototype is therefore not an empirical neural benchmark and should not be read as one. It is a finite declared-regime implementation of the certificate discipline. The paper also does not provide a complete theory of learned implicit observation maps. It does not provide an empirical validation of the boundary-revision hypothesis advanced in Section 9 as an account of human cognition; that hypothesis is offered as philosophical motivation and is explicitly not claimed as an established result. It does not provide a complete stochastic admissibility calculus, only the sketch given in Section 13. It does not provide a full engineering design for autonomous regime reconstruction; Section 11 identifies the required architectural separation without specifying every mechanism by which reconstruction authority should be exercised in a deployed system.

Several open problems follow directly from these gaps and are worth stating individually rather than leaving implicit. The first is the problem of implicit learned maps: how can one identify the effective observation map of a learned model whose representations are distributed across high-dimensional weights and activations, given that no such map is declared anywhere in the system's architecture? The second is the problem of admissibility witnesses: what would count as an inspectable witness that a learned representation preserves the distinctions required by a target predicate, in a form analogous to the residue certificate of Section 5 but applicable to systems for which no finite regional structure is given in advance? The third is the problem of inadequacy signals: beyond the candidate classes listed in Section 11, what signals reliably indicate that an observation regime has ceased to preserve the relevant distinctions, and how can such signals be distinguished from ordinary noise within a regime that remains adequate? The fourth is the problem of controlled regime revision: when should closure authority revise the operative regime rather than only reject a proposal made within it, and what governs the transition between these two responses? The fifth is the problem of governance: closure authority in consequential artificial intelligence systems is not only technical. It is institutional. Some governed process, and not only a piece of software, must hold the authority to declare a regime inadequate and to oversee its revision, and this paper does not resolve what that process should be.

15 Conclusion: From Confidence Scores to Admissibility Witnesses

Optimisation presupposes admissibility. Where the observational regime under which a system acts collapses distinctions required by the target predicate, further optimisation cannot restore the warrant that judgement requires, regardless of how much further optimisation is applied or how it is shaped.

The paper has developed five concrete supports for this claim. It gives a finite worked obstruction, in which local adequacy is shown, by explicit calculation rather than by analogy, to fail to lift to global warrant. It gives an executable finite prototype, in which a cyclic diagnostic system emits certificates distinguishing coherence failure, global admissibility, and warrant debt. It gives a reward-hacking schema, written as an explicit triple, in which the same obstruction structure is made visible in a setting of direct relevance to reinforcement learning. It gives a benchmark and proxy-failure schema, written as a second explicit triple, in which the same structure is made visible in a setting of direct relevance to benchmark-driven development. And it has given a closure architecture, separating productive search from closure authority and introducing warrant debt as the residual quantity that closure authority is responsible for detecting.

Reliable artificial intelligence should not stop at answers with confidence scores, since a confidence score is still a function of the regime already in force. The next requirement is an admissibility witness: a checkable indication that the regime under which the answer is offered preserves the distinctions on which the answer depends, or a signal that this regime has failed.

Code Availability

A finite prototype, CycleDebt, is provided in the public repository `https://github.com/dhwcmoore/cycledebt`. The repository contains a Python diagnostic engine, generated JSON certificates, an independent OCaml/Zarith verifier, a Rocq certificate core, and a HELICS co-simulation demonstration. The Rocq core proves the main soundness conditions, with the remaining linear-solve component checked exactly by the Python and OCaml layers. The repository is supplied as supporting material for the finite declared-regime claims made in

Section 5 and Section 11.6. It is not claimed to solve the open implicit-map problem for learned neural systems.

A companion Rocq/OCaml formalisation of the structural admissibility framework is available at `https://github.com/dhwcmoore/structural-admissibility`. It mechanises the factorisation criterion, safety-relevant collapse, warrant-debt witnesses, refinement and restriction repairs, runtime rupture certificates, and declared-regime case studies for physical sensing, distributed consensus, and GPU memory consistency. This second repository supports the formal and architectural claims of the paper, while the repository hosting the CycleDebt prototype supports the executable obstruction certificate used in Section 11.6. Its declared assumptions are documented in the repository rather than hidden in the prose of this paper.

A Quotient Descent and Factorisation Proofs

This appendix presents, side by side, two formal expressions of the closure obligation referenced in the body of the paper. The first is the admissibility factorisation, a statement about quotient maps in the algebraic setting. The second is the weakest-precondition semantics of Dijkstra's non-deterministic guarded conditional, its operational counterpart. Each has been studied in its own literature, and neither requires the other for its statement. Presenting them together makes explicit that the closure obligation, in either formulation, is the demand for uniformity of a property over a partition, where the partition is supplied either by an observational regime or by a guard system.

A.1 Admissibility as quotient factorisation

Let X be a set of states and O a set of observations available to the system. Let $M : X \to O$ be the observation map. The kernel equivalence of M is the relation $x \sim_M y \iff M(x) = M(y)$. The relation $\sim_M$ induces a partition of X whose blocks are the fibres of M. Write $X/{\sim_M}$ for the resulting quotient set, and let $\pi : X \to X/{\sim_M}$ be the canonical projection. The map M itself factors as $M = \widetilde{M} \circ \pi$, where $\widetilde{M} : X/{\sim_M} \to O$ is injective. The image $M(X) \subseteq O$ is therefore in bijection with $X/{\sim_M}$. This is the algebraic content of the observation regime: up to relabelling of observations, the regime is the partition structure $\sim_M$.

Proposition 3 (Admissibility factorisation). *Let $\Phi : X \to \{0, 1\}$ be a predicate. The following are mutually equivalent: that Φ is admissible with respect to M in the sense of Definition 1; that Φ is constant on each block of the partition $\sim_M$; that there exists a unique map $\hat{\Phi} : X/{\sim_M} \to \{0, 1\}$ satisfying $\Phi = \hat{\Phi} \circ \pi$; and that there exists a map $\bar{\Phi} : O \to \{0, 1\}$ satisfying $\Phi = \bar{\Phi} \circ M$.*

Proof. That admissibility is equivalent to constancy on blocks restates Definition 1 in partition language. That constancy on blocks yields a unique factorisation through π is the universal property of the quotient set, by which any function on X respecting $\sim_M$ descends uniquely along the projection. The existence of $\bar{\Phi}$ follows by composing $\hat{\Phi}$ with $\widetilde{M}^{-1}$ on the image $M(X)$ and extending arbitrarily on $O \setminus M(X)$. The converse, that any factorisation through M yields admissibility, is immediate from the definitions. □

Proposition 4 (Apparent randomness under non-injective observation). *Let $M : X \to O$ be an observation map and let an agent's behaviour be determined by a function $f : X \to Y$ whose values depend on distinctions not preserved by M, in the sense that there exist $x, y \in X$ with $M(x) = M(y)$ but $f(x) \neq f(y)$. Then there is no function $g : O \to Y$ satisfying $f = g \circ M$, and any external observer modelling the agent as a function of O alone is forced to record f as a non-functional, hence stochastic-looking, relation on O.*

Proof. Suppose, for contradiction, that $f = g \circ M$ for some $g : O \to Y$. Then $M(x) = M(y)$ entails $g(M(x)) = g(M(y))$, and hence $f(x) = f(y)$, contradicting the hypothesis. No such g

exists. The fibre $M^{-1}(M(x))$ accordingly carries multiple values of f, and an evaluator restricted to functions of M assigns multiple values of Y to a single observation in O, which is the structural form of a non-deterministic, and hence stochastic-looking, relation on O. □

A.2 The weakest-precondition calculus for guarded conditionals

The weakest-precondition calculus belongs to the axiomatic tradition of program semantics inaugurated by Hoare (Hoare, 1969). Following Dijkstra, consider the non-deterministic conditional

$$\text{IF} \equiv \textbf{if } B_1 \to S_1 \ [] \ \cdots \ [] \ B_n \to S_n \textbf{ fi},$$

in which each B_i is a Boolean predicate over the state space and each S_i is a command (Dijkstra, 1975, 1976). Let $BB \equiv B_1 \vee \cdots \vee B_n$. The weakest-precondition semantics of IF with respect to a postcondition R is

$$\text{wp}(\text{IF}, R) \equiv BB \wedge \bigwedge_{i=1}^{n} \big(B_i \Rightarrow \text{wp}(S_i, R)\big).$$

The first conjunct ensures that some guard holds at the initial state, so that IF does not abort. The second conjunct ensures that, for every guard that holds, the corresponding branch establishes the postcondition. The proof obligation is universal over the branches whose guards are true, rather than existential.

Define the equivalence relation on the state space by

$$x \sim_{\text{IF}} y \iff \forall i \in \{1, \ldots, n\}.\ B_i(x) = B_i(y),$$

so that two states are equivalent when the same set of guards is true at each. The relation $\sim_{\text{IF}}$ partitions the state space according to which subset of guards is active. Within a single block of this partition the set of available branches is fixed, and the weakest-precondition obligation says that R is established by every such branch: uniformity of the postcondition over the partition, mediated by the available branches.

A.3 Closure as uniformity over a partition

The two formulations exhibit the same structural pattern. A representational structure, whether an observation regime or a guard system, partitions the state space. A property of interest, whether a static predicate or a postcondition under execution, is required to be invariant across the partition cells in the appropriate sense for the setting. Where the property is not so invariant, the representational structure has failed to preserve the distinction the property was attempting to draw, and the closure obligation has failed. The two presentations are not isomorphic, the first concerning a single observation map and a single predicate and the second a compound program acting over a state space, but both exhibit the closure obligation in its general form: that a property of interest be uniform across the partition imposed by the surrounding structure.

B Finite Regional Obstruction Details

This appendix gives the technical detail of the finite obstruction example introduced in Section 5.

B.1 The regional system

Let a finite cover $\{U_i\}_{i=1}^{n}$ of a domain be given, together with the pairwise overlaps $U_{ij} = U_i \cap U_j$ and, where relevant, the triple overlaps $U_{ijk} = U_i \cap U_j \cap U_k$. On each U_i a local assignment a_i is given, and on each U_{ij} a comparison of the restrictions of a_i and a_j to the overlap is available. A global assembly is a single assignment a on the whole domain whose restriction to each U_i agrees with a_i.

B.2 A Čech-style complex of corrections

Following the standard Čech construction adapted to this setting, let C^0 be the group of local correction data on the regions U_i, let C^1 be the group of comparison data on the overlaps U_{ij}, and let C^2 be the group of coherence data on the triple overlaps U_{ijk}. Define a local correction operator

$$\delta_0 : C^0 \to C^1$$

sending a tuple of local corrections to the discrepancy it induces on each overlap, and a coherence operator

$$\delta_1 : C^1 \to C^2$$

sending a tuple of overlap discrepancies to the alternating sum of its values around each triple overlap, in the usual Čech pattern. The composite $\delta_1 \circ \delta_0$ vanishes identically, by the same telescoping cancellation familiar from the Čech complex of any cover, so that $\operatorname{im} \delta_0 \subseteq \ker \delta_1$.

B.3 The obstruction

Because $\operatorname{im} \delta_0 \subseteq \ker \delta_1$, every residue arising from local data that is at least pairwise consistent in the appropriate sense satisfies $\delta_1 r_x = 0$. The obstruction case is a configuration for which

$$\delta_1 r_x = 0 \quad \text{but} \quad r_x \notin \operatorname{im} \delta_0.$$

Such a residue defines a non-trivial class $[r_x]$ in the quotient $\ker \delta_1 / \operatorname{im} \delta_0$, the first Čech cohomology of the regional system with coefficients in the relevant correction sheaf. The class $[r_x] \neq 0$ is the formal statement of warrant debt for the configuration x: the residue is internally coherent across every triple overlap, so no local inconsistency can be pointed to, and yet no choice of local correction cancels it, so no global assembly exists. Because C^0, C^1, and C^2 are finite-dimensional in the finite case under discussion, the class $[r_x]$ is computable by finite linear algebra, and the obstruction is therefore checked, not asserted.

B.4 A worked instance and its general solution

We now give the example used in Section 5 in full, including the exact characterisation of which residues obstruct and which do not, so that the single numerical case given there is shown to be representative rather than special. Take the cyclic cover of four regions U_1, U_2, U_3, U_4 described in Section 5, with consecutive overlaps $U_{12}, U_{23}, U_{34}, U_{41}$ and no others, and no triple overlaps. Then $C^0 \cong \mathbb{Q}^4$ with coordinates (c_1, c_2, c_3, c_4), $C^1 \cong \mathbb{Q}^4$ with coordinates $(a_{12}, a_{23}, a_{34}, a_{41})$, and $C^2 = 0$.

The local correction operator, written as a matrix acting on column vectors, is

$$\delta_0 = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \\ 1 & 0 & 0 & -1 \end{pmatrix}.$$

Its kernel consists of the constant vectors (c, c, c, c) for $c \in \mathbb{Q}$, a one-dimensional subspace, so by the rank-nullity theorem δ_0 has rank $4 - 1 = 3$ and its image is a 3-dimensional subspace of $C^1 \cong \mathbb{Q}^4$. Summing the four rows of δ_0 gives the zero row, since each c_i appears once with coefficient $+1$ and once with coefficient -1 across the four equations; consequently every vector in $\operatorname{im} \delta_0$ has coordinate sum zero. Since this condition already cuts out a 3-dimensional subspace and $\operatorname{im} \delta_0$ is itself 3-dimensional, the two subspaces coincide:

$$\operatorname{im} \delta_0 = \{(a_{12}, a_{23}, a_{34}, a_{41}) \in \mathbb{Q}^4 : a_{12} + a_{23} + a_{34} + a_{41} = 0\}.$$

Because $C^2 = 0$, every $r \in C^1$ satisfies $\delta_1 r = 0$ trivially, so the coherence check imposes no constraint here, exactly as for the Čech cohomology of a circle, where H^2 vanishes and the entire obstruction is carried by H^1. It follows that a residue $r = (a_{12}, a_{23}, a_{34}, a_{41})$ obstructs global assembly, that is, $[r] \neq 0$ in $C^1/\operatorname{im}\delta_0$, when its coordinate sum is non-zero:

$$[r] \neq 0 \iff a_{12} + a_{23} + a_{34} + a_{41} \neq 0,$$

and in that case the quotient $C^1/\operatorname{im}\delta_0 \cong \mathbb{Q}$ is represented exactly by this sum. The residue $r = (1, 1, 1, 1)$ used in Section 5 has coordinate sum $4 \neq 0$ and is therefore a representative obstruction, but so is any residue whose four coordinates do not sum to zero, while any residue whose coordinates do sum to zero, for instance $(1, 1, 1, -3)$ or $(2, -2, 1, -1)$, is exact and admits an explicit local correction recoverable by solving the system directly. This gives the regional structure of Section 5 a complete, checkable classification of its obstructions, rather than a single isolated example.

B.5 The residue, in general

Given a regional configuration x over an arbitrary finite cover, the discrepancies observed on the pairwise overlaps U_{ij} define an element $r_x \in C^1$, the residue of x. The configuration x admits a global assembly, that is, $\Phi(x) = 1$, when there is a choice of local corrections $c \in C^0$ with $\delta_0(c) = r_x$, that is, when $r_x \in \operatorname{im}\delta_0$. The cyclic example above is the case $n = 4$ of this general construction, chosen because its obstruction group $C^1/\operatorname{im}\delta_0$ is one-dimensional and admits the closed-form description given above; covers with non-empty triple overlaps introduce a non-trivial δ_1 and a correspondingly richer obstruction theory, which the general definitions above accommodate without modification.

B.6 Interpretation

The computation above should be read together with the distinction introduced in Section 5 between the weak local regime M_{loc} and the richer certificate regime M_{cert}. The weak local regime records the local coherence status of the profile: whether the available local checks report a contradiction, for instance a non-zero triple-overlap defect detected by the failure of the check $\delta_1 r = 0$. It does not, by itself, record the obstruction class $[r_x]$ or perform the exactness calculation that determines whether $r_x \in \operatorname{im}\delta_0$.

The certificate regime, by contrast, records the data needed for that calculation,

$$M_{\text{cert}}(x) = (\delta_0, r_x),$$

which in the four-cycle example reduces to the finite check

$$a_{12} + a_{23} + a_{34} + a_{41} = 0.$$

The same regional configuration can accordingly be treated in two ways. Relative to M_{loc}, the global-coherence predicate need not be decidable, since local coherence does not by itself determine whether the residue is exact. Relative to M_{cert}, the predicate is decidable by a finite verifier, as Proposition 3 guarantees once δ_0 and r_x are both supplied. The point of the example is therefore not that the full residue data are insufficient to decide Φ. The point is that local coherence alone is insufficient, and that closure authority consists in supplying and independently checking the additional certificate data that make the global predicate decidable.

C Certificate Schema and Verifier Architecture

This appendix describes the certificate schema underlying the architectural proposal of Section 11, abstracted from the finite case of Appendix B and matching Algorithm 1 in the main text.

A certificate, in the finite setting, consists of the regional data $\{a_i\}$, the overlap discrepancies that make up the residue r_x, and a record of the coherence check $\delta_1 r_x = 0$ on each triple overlap. The checker, given a certificate, performs two finite computations: it confirms that $\delta_1 r_x = 0$, and it searches the finite-dimensional space C^0 for a correction c with $\delta_0(c) = r_x$. A configuration is certified globally admissible when such a c is found. A configuration is certified as carrying warrant debt when the coherence check passes, $\delta_1 r_x = 0$, but the search for a correction provably fails, which in the finite-dimensional linear setting is itself a finite, checkable computation rather than an open-ended search. The role of an independent verifier is to repeat this checking procedure using an implementation separate from whatever process proposed the original configuration, so that the certification does not depend on the proposer's own report of its result. Where the underlying setting is not finite, for instance where the regional structure or the relevant correction space is infinite-dimensional or only implicitly specified, the search for a correction c may not terminate, and establishing warrant debt then requires an independent non-existence argument rather than an exhaustive search; we do not attempt to characterise such arguments in general, and the certificate schema given here should accordingly be read as a finite prototype rather than a general-purpose verification procedure.

C.1 The schema applied to the worked instance

On the cyclic four-region example of Appendix B, the schema is fully explicit. A certificate for the configuration carrying residue $r = (1, 1, 1, 1)$ consists of the four overlap discrepancies themselves, together with the empty record of triple-overlap data, since $C^2 = 0$ for this cover and the coherence check $\delta_1 r = 0$ holds without further computation. A checker presented with this certificate need perform only one further step to certify warrant debt: sum the four coordinates of r and confirm that the sum, here 4, is non-zero, which by the closed-form description of $\operatorname{im} \delta_0$ established in Appendix B is necessary and sufficient for $r \notin \operatorname{im} \delta_0$. An independent verifier, given only the four numbers $(1, 1, 1, 1)$ and the matrix δ_0, can reproduce this conclusion without reference to whatever process proposed the original regional assignments, which is the property the architecture of Section 11 requires of closure authority generally: the certificate and the check together are sufficient to settle the question, and neither party need trust the other's report of the result.

D Sketch of Stochastic Generalisation

This appendix records, briefly, the technical sketch underlying Section 13. Let $K(o \mid x)$ be a stochastic observation kernel on $X \times O$. The deterministic case of Section 4 is recovered when $K(\cdot \mid x)$ is a point mass at $M(x)$ for some function $M : X \to O$.

The exact stochastic analogue of admissibility, stated first in Section 13, is

$$K(\cdot \mid x) = K(\cdot \mid y) \implies \Phi(x) = \Phi(y),$$

which reduces immediately to Definition 1 once K degenerates to a point mass at $M(x)$, since $K(\cdot \mid x) = K(\cdot \mid y)$ then becomes exactly $M(x) = M(y)$.

The relaxed, decision-theoretic version weakens equality of Φ to equality of the conditional probability of Φ, and can be stated equivalently as the existence of a function $\bar{\Phi}$ on the space of observational laws such that

$$P(\Phi = 1 \mid X = x) = \bar{\Phi}\big(K(\cdot \mid x)\big)$$

for all $x \in X$, that is, the conditional law of Φ given $X = x$ depends on x only through the induced distribution over observations. This is again a direct stochastic analogue of the factorisation condition of Proposition 3, with the deterministic equality $M(x) = M(y) \Rightarrow \Phi(x) = \Phi(y)$ replaced by agreement of conditional laws rather than agreement of point observations. One caution is required in comparing the two conditions. When the target remains, as in the deterministic theory, a function $\Phi : X \to \{0, 1\}$ of the state, the conditional probability $P(\Phi = 1 \mid X = x)$ equals $\Phi(x)$, and the relaxed condition coincides with the exact one. The relaxed condition is strictly weaker only when the target is modelled as a random variable jointly distributed with X rather than as a function of it, in which case it is implied by, but does not imply, the exact condition stated above. The connection to classical sufficiency is direct: a sufficient statistic for an inferential target is exactly an observation regime, in this relaxed sense, that is admissible for predicates concerning that target. We do not develop the stochastic fixed-regime limit, the natural analogue of Proposition 1 in this setting, in this paper, and we record it here only as a next step in the research programme this paper proposes.

Declarations

Ethical approval

This article does not contain any studies involving human participants or animals performed by the author. The manuscript is a theoretical and computational methods paper using formal examples, mathematical arguments, and reproducible computational artefacts.

Consent to participate

Not applicable. No human participants were involved in this study.

Consent to publish

Not applicable with respect to participant data, since no human participants were involved. The author consents to publication of the manuscript and its original scholarly content.

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